

Importance Sampling for Deterministic Uncertainty when Estimating QoT in Autonomous Optical Networks

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Abstract *We investigate importance sampling to significantly improve computational efficiency when assessing the probability of lightpath failure in complex scenarios. Results pave the way towards risk-informed decision-making in self-driven optical networks. ©2026 The Author(s)*

Introduction

Optical networks are increasingly needed to operate close to their physical limits [1], to maximize spectral efficiency and reduce the cost-per-bit. This is limited in practice by uncertainties on key inputs and QoT model accuracy [2]. Combined with worst-case design [3-4], this results in over-dimensioned systems with excessive margins [5].

Beyond design, in autonomous networks, decision-making would rely on QoT estimation for a wide variety of tasks. Here, a worst-case approach would inevitably result in crippled automation, the safest choice without proper risk assessment being to do nothing. Thus, there is a need for proper statistical evaluation of the risks involved in contemplated actions [6], which can be summarized as deterministic uncertainty.

When actions involve optical lightpaths, deterministic uncertainty comes down to evaluating the failure probability, i.e. the probability of the lightpath QoT to be below FEC limit. In this context, worst-case design assumes the most pessimistic combination of inputs within a predefined range. While this widely adopted strategy is meant to achieve permanent zero-risk, it fundamentally relies on assumptions which are not necessarily verified, as per the countless observations of unexpected failures in field trial data [7]. Thus, not only the current approach cannot assess the gain from over-engineering optical networks, it also fails at guaranteeing zero-risk.

In contrast, statistical design [8] allows to find the best margin-availability trade-off [5] by assuming QoT inputs to be random variables. In previous work [9], we assumed those to be normally distributed. This significantly simplifies calculations due to the existence of closed-form formulas for propagating the uncertainty from inputs to QoT. However, in autonomous, digital-twin enabled networks [10], complexity lies in that input knowledge comes in multiple forms: datasheet, pre-deployment characterization, individual calibrations, direct measurements, input refinement, etc. All with distributions that can largely deviate from gaussian. This is especially problematic considering the normal approximation specifically

impacts the QoT distribution's tail from which the risk of failure is evaluated.

In such complex yet topical scenarios, the Monte-Carlo (MC) method is ideal in that it only requires minimal assumptions on inputs, for instance leveraging the principal of maximal entropy for a conservative risk assessment [11]. It consists in repeatedly evaluating lightpath QoT under random realizations of its inputs. However, each QoT evaluation may involve computationally expensive models (e.g. IGN, EGN, split-step simulations), making its direct use, i.e. crude Monte-Carlo (CMC), prohibitively slow.

To address this, we investigate in this paper the use of importance sampling (IS) [12-15] to estimate the risk of insufficient QoT much faster than with crude Monte-Carlo (CMC). We first present a simple methodology to apply IS to QoT estimation, using a typical optical lightpath for illustration. Then, we apply IS to a complex optical restoration scenario for estimating the risk associated with switching to a backup path.

Methodology for Importance Sampling (IS)

The core principle of IS is to bias Monte-Carlo simulations to explore combinations of inputs that are more likely to lead to failures than default ones, allowing to significantly reduce the number of trials required to reach a given accuracy when estimating the lightpath failure probability P_f .

For hands-on explanation, let us consider the case of a single coherent optical lightpath whose QoT is characterized by the SNR, i.e. the signal-to-noise ratio of the complex, polmux digital signal at reception, after DSP and before decision. The SNR may be calculated from any given QoT model as a function of the lightpath inputs.

Step 1 consists in stating the problem, starting with the definition of the limit state function (LSF), which by convention is negative in case of failures, and positive otherwise. For a lightpath, we simply have $LSF = SNR - SNR_{FEC}$, where SNR_{FEC} is the FEC limit. The LSF presents as a function of all inputs relevant to the estimation of the lightpath's SNR, which can be either known, or

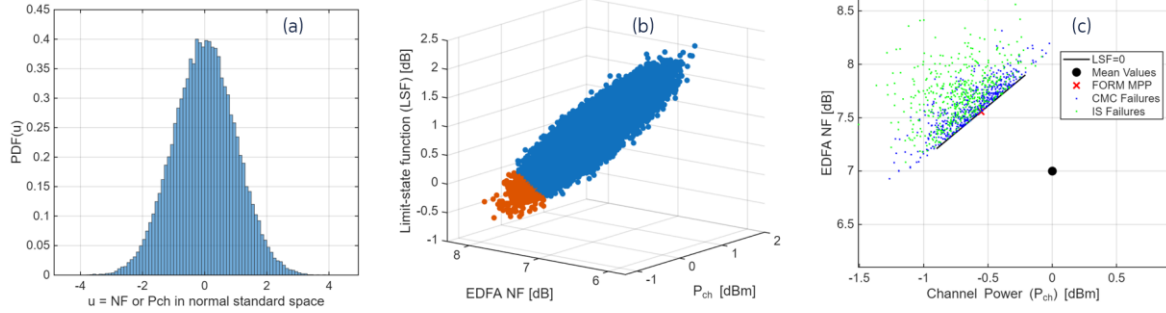


Fig. 1: (a) Distribution for the random inputs in the normal standard space or u-space. (b) Evaluation of the limit state function (LSF) as a function of the random inputs, here the EDFA noise figure (NF) and channel launch power. Red dots correspond to input combinations leading to failures. (c) Summary illustration of the importance sampling (IS) method.

random. Step 1 also consists in defining which inputs are random, and how they are distributed.

Step 2 consists in using crude Monte-Carlo (CMC) to estimate the LSF on the random parameter space. Optional, it provides a first estimation of the failure probability, which can be useful to select a suitable amount of IS trials during subsequent steps.

Step 3 consists in finding the central bias for the subsequent sampling – i.e. the most probable point (MPP) – using for instance a first order regression method (FORM). This is done first by converting the parameter space into a normal standard space or u-space. Then, a constrained optimization algorithm is used to find u_{MPP} that minimizes $\|u\|^2$ while verifying $LSF_u(u)=0$. The MPP defines the mean of the random inputs explored during the last step.

Step 4 ends with the IS-based estimation of the failure probability $P_f = P(LSF < 0)$. We first draw N_{IS} sets of inputs centered on the MPP and calculate the corresponding LSF values. The bias applied for sampling needs to be corrected to deduce an unbiased estimation of P_f . This is done by applying importance weights as the ratio between the initial and the biased PDFs.

We illustrate these steps on a PDM-16QAM lightpath at 100Gb/s over eleven 80-km-spans of standard single mode fiber (SSMF) with 0.2 dB/km loss. We further assume that the channel launch power P_{ch} is normally distributed around 0 dBm with 0.3 dB of standard deviation, while the EDFA noise figures NF, assumed uniform in the lightpath, are normally distributed around 7 dB, also with 0.3 dB standard deviation. With $SNR_{trx} = 20$ dB, the SNR is estimated from:

$$SNR^{-1} = SNR_{ASE}^{-1} + SNR_{NLI}^{-1} + SNR_{trx}^{-1}$$

To simplify calculations, we arbitrarily assume that the launch power always matches the non-linear threshold, fixing the ratio between ASE and NLI noise. The various steps involved in the IS methodology are illustrated in Fig. 1.

In Fig. 1a, we plot the sampled PDF for either NF or P_{ch} in the normal standard space (zero-mean, unitary standard deviation). In Fig. 1b, we

plot the corresponding values of the LSF evaluated in Step 2 using crude Monte-Carlo (CMC) with $N_{CMC}=10^5$ trials. The separation between blue dots (normal) and red dots (failures) marks the $LSF=0$ contour where the MPP is found during step 3. This MPP is further represented in the 2D parameter space in Fig. 1c as a red cross, with all CMC and IS samples corresponding to failures, with $N_{IS}=10^3$ trials. With $N_{CMC}=100N_{IS}$, there are similar numbers of failures for IS (Step 4) and CMC (Step 2), illustrating how IS can achieve a similar accuracy for P_f estimation despite the much lower number of trials.

The gain in terms of reduction of required number of trials is further explored in Fig. 2, where we plot the RMSE of the estimated failure probability P_f for CMC and IS methods, as functions of the number of trials N . Both IS and CMC show the same trends, but IS systematically achieves 10-times better accuracy than CMC for the same number of trials, converting into an average 100x reduction of required trials to achieve a given accuracy, following the expected behavior that accuracy scales with N^2 [12].

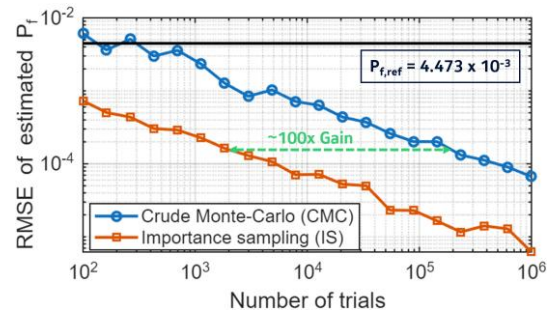


Fig. 2: 100x reduction of required trials from crude Monte-Carlo (CMC) to importance sampling (IS) to achieve the same accuracy on the risk P_f of LP failure

Application to an optical restoration use case

Here, we apply the IS methodology to a significantly more complex use case, consisting in estimating the risk of switching from an active LP to a backup LP in an arbitrary optical network where QoT input knowledge comes from a variety of sources, as illustrated in Fig. 3a.

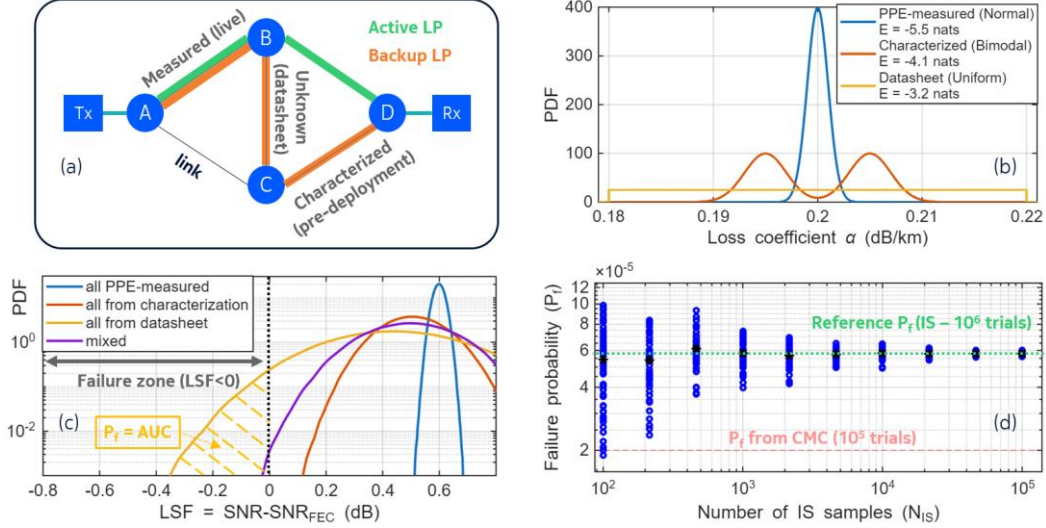


Fig. 3: (a) Considered topology with active and backup lightpaths (LP). (b) Statistical distributions describing span losses depending on the link. (c) CMC-based risk evaluation for different input knowledge scenarios. (d) Convergence of the IS-based failure probability estimation with an increasing number of trials.

For simplicity, we assume all inputs to be perfectly known except the loss coefficients α of each span, treated as independent random variables. We further assume the active LP is equipped with power profile estimation (PPE) monitoring [16], which results in live measurements of α values for the four spans of link A-B. The error for PPE-measured α values is assumed to follow a normal distribution with 0.001 dB/km standard deviation, as illustrated in Fig.3b. In contrast, the four spans of link C-D envisaged for the backup LP are not monitored by PPE. Yet, it is known after a pre-deployment characterization of a significant number of spans that any deployed span of that link would have a α value following the bimodal distribution plotted in Fig.3b, approximately matching that of an undisclosed, real-life result. Finally, the four spans of link B-D have not been PPE-monitored nor characterized pre-deployment. The only available knowledge is from the fiber manufacturer datasheet, stating that a span's α value is anywhere between 0.18 and 0.22 dB/km. Here, we apply the law of maximal entropy to convert this datasheet information into an assumed uniform distribution. This represents the most conservative assessment of the uncertainty when nothing is known but the extreme values, strongly contrasting with the worst-case paradigm which systematically assuming the worst α for each span despite the infinitesimal probability of such event.

For comparing the impact of the various types of knowledge on the risk of failure for the backup LP, we plot in Fig.3c the CMC-obtained PDFs of the limit state function (LSF), i.e. the SNR margin, in three alternative scenarios. The first, corresponding to the 12 spans measured with PPE, naturally presents the lowest SNR uncertainty

and the lowest risk, too small to be estimated. In contrast, the weaker knowledge that all spans are from a batch characterized before deployment significantly increases the risk, but less than the weakest assumption deriving from the datasheet. We illustrate on this third scenario the calculation of P_f as the area under the curve (AUC) for $LSF < 0$. The actual backup LP, described previously, corresponds to the “mixed” scenario in Fig.3c, with the failure probability $P_{f,ref} = 5.5 \times 10^{-5}$, significantly driven by the large uncertainty from the “unknown” spans of link B-C.

Finally, we apply the IS methodology to the backup LP in the mixed scenario. The most critical step is step 3, where each type of distribution requires a specific transformation of the initial input space to the u-space. In Fig.3d, we plot the results of 20 runs of failure probability evaluations with various amounts of IS trials N_{IS} , from 10^2 to 10^5 . For instance, $N_{IS}=100$ results in relatively large variations of the estimated P_f depending on the run, hence a large RMSE. The RMSE however improves by increasing N_{IS} , as expected, with a progressive convergence toward $P_{f,ref}$. In comparison, the estimated P_f for one single CMC run with $N_{CMC}=10^5$ is $2 \cdot 10^{-5}$, corresponding to a much higher RMSE than IS for the same number of trials.

Conclusions

We investigated importance sampling (IS) for proper assessment of failure risk, to achieve deterministic uncertainty in automated optical networks. We showed that IS typically reduces required simulation trials by 100 compared to crude Monte-Carlo (CMC). We further showed that IS successfully applies to complex yet topical and future-proof network scenarios.

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